

MATHEMATICAL METHODS OF PHYSICS ***(MMP-1)***

**Past Paper First Annual-2022 Fall
Solution**





UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Roll No. 48178

Subject: Mathematical Methods of Physics - I

Course Code: PHYS-3501

Time: 3 Hrs.

Marks: 60

Q.1. Give short answers of the following:

(6x5=30)

1. Prove that the operator $\mathcal{L} = -\frac{d}{dx}\left(p_0(x)\frac{d}{dx}\right) - p_2(x)$ is Hermitian over the range $[a, b]$ under boundary condition $\left[y_i^* p_0 y_j'\right]_{x=a}^{x=b}$
2. By using mathematical induction, prove that $J_n(x) = (-1)^n x^n \left(\frac{1}{x} \frac{d}{dx}\right)^n J_0(x)$
3. Put the following equation into Sturm-Liouville (SL) form:

$$(1 - x^2)y''' - xy' + v^2y = 0$$
4. Find Fourier transform of Gaussian distribution function $f(x) = e^{-at^2}$ with $a > 0$.
5. Verify the beta function identity $B(a, b) = B(a+1, b) + B(a, b+1)$
6. By using the generating function for Hermite polynomial, $e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$ show that $H'_n(x) = 2nH_{n-1}(x)$

Q.2 Answers the following questions.

(5x6=30)

1. Find the Green's function for the equation $-\frac{d^2y}{dx^2} - \frac{y}{4} = f(x)$ with boundary conditions $y(0) = y(\pi) = 0$.
2. Using the Rodrigues formula $P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx}\right)^n (x^2 - 1)^n$, show that $P_n(x)$ are orthogonal and that $\int_{-1}^1 P_n(x) P_n(x) dx = \frac{2}{2n+1}$
3. Show that $\Gamma(1+n)\Gamma\left(n+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n}} \Gamma(2n+1)$ where n is an integer.
4. By directly differentiating $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin\theta) d\theta$ show that the recurrence relation $2J'_n(x) = \frac{1}{2}[J_{n-1}(x) - J_{n+1}(x)]$
5. The function $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$ is a symmetrical finite step function; Find the Fourier cosine transform of $f(x)$.

MMP-I (Past Paper - 2022 (fall))

Q#1 Give short answers:

1- Prove that the operator $\mathcal{L} = -\frac{d}{dx} \left(p_0(x) \frac{d}{dx} \right) - p_2(x)$ is Hermitian over the range $[a, b]$ under boundary conditions $[y_i^* p_0 y_j']_{x=a}^{x=b}$

Step 1

Find the adjoint of the given operator.

Substitute $p_0(x)$ and $p_2(x)$ with their complex conjugates $p_0^*(x)$ and $p_2^*(x)$, respectively, in the given operator.

$$-\frac{d}{dx} \left(p_0(x) \frac{d}{dx} \right) - p_2(x)$$

$$-\frac{d}{dx} \left(p_0^*(x) \frac{d}{dx} \right) - p_2^*(x)$$

Rewrite the operator.

$$-\frac{d}{dx} \left(p_0^*(x) \frac{d}{dx} \right) - p_2^*(x)$$

Step 2



Compare the obtained operator

$$-\frac{d}{dx} \left(p_0^*(x) \frac{d}{dx} \right) - p_2^*(x) \text{ with the}$$

definition of the adjoint operator

$$A^* = -\frac{d}{dx} \left(p_0^*(x) \frac{d}{dx} \right) - p_2^*(x).$$

Notice that the obtained operator is the same as the adjoint operator.

Therefore, the given operator is symmetric.

Step 3

Since the given operator is symmetric, it is Hermitian.



By using mathematical induction, prove that $J_n(x) = (-1)^n x^n \left(\frac{1}{x} \frac{d}{dx}\right)^n J_0(x)$

base
Verification this case for $n=0$: the right side of equation becomes
 $(-1)^0 x^0 \left(\frac{1}{x} \frac{d}{dx}\right)^0 J_0(x)$
 $J_0(x)$ R.H.S $\Rightarrow$ L.H.S $\Rightarrow J_n(x)$ when $n=0 \Rightarrow J_0(x)$
 $J_0(x) = J_0(x)$

The base case holds true.
 Assuming the statement is true for $n=k$: $k = \text{arbitrary +ve integer}$
 then: $J_k(x) = (-1)^k x^k \left(\frac{1}{x} \frac{d}{dx}\right)^k J_0(x)$

For $n=k+1$
 $J_{k+1}(x) = (-1)^{k+1} x^{k+1} \left(\frac{1}{x} \frac{d}{dx}\right)^{k+1} J_0(x)$

Conclusion
 Statement holds true for $n=0, k, k+1$.
 It means it is true for all non-negative integers n .

3- Put the following equation in SL form:
 $(1-x^2)y'' - xy' + v^2y = 0$

Solution
 As; $\frac{d}{dx} \left[P(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)]y = 0$
 As given; $(1-x^2)y'' - xy' + v^2y = 0$

$$y'' - \frac{x}{1-x^2} y' + \frac{v^2}{1-x^2} y = 0 \rightarrow \textcircled{A}$$

Integrating Factor I.F $e^{\int P(x) dx} = e^{-\int \frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \int \frac{2x}{1-x^2} dx}$

$$= e^{\frac{1}{2} \ln(1-x^2)} = e^{\ln(1-x^2)^{1/2}} = (1-x^2)^{1/2}$$

$$\text{I.F} = \sqrt{1-x^2}$$

Multiplying I.F with eq (A)

$$\sqrt{1-x^2} y'' - \frac{x}{\sqrt{1-x^2}} y' + \frac{v^2}{(1-x^2)^{1/2}} y = 0$$

$\frac{d}{dx} \left[\sqrt{1-x^2} \frac{dy}{dx} \right] + \frac{v^2}{(1-x^2)^{1/2}} y = 0$	S.L form
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$$P(x) = (1-x^2)^{1/2} ; q(x) = 0 \quad r(x) = \frac{1}{(1-x^2)^{1/2}}$$

4- Find Fourier transform of Gaussian distribution function

$$f(x) = e^{-at^2} \text{ with } a > 0.$$

Solution

$$F(s) = \int_{-\infty}^{\infty} e^{-at^2} e^{ist} dt = \int_{-\infty}^{\infty} e^{-a(t^2 - \frac{ist}{a})} dt$$

$$= \int_{-\infty}^{\infty} e^{-a \left[\left(t - \frac{is}{2a}\right)^2 - \left(\frac{is}{2a}\right)^2 \right]} dt = e^{a \left(\frac{is}{2a}\right)^2} \int_{-\infty}^{\infty} e^{-a \left(t - \frac{is}{2a}\right)^2} dt$$

$$\text{Let } y^2 = a \left(t - \frac{is}{2a}\right)^2 \Rightarrow dy = \sqrt{a} dt$$

$$= e^{a \left(\frac{is}{2a}\right)^2} \int_{-\infty}^{\infty} e^{-y^2} \frac{dy}{\sqrt{a}}$$

$$= e^{a \left(\frac{is}{2a}\right)^2} \int_{-\infty}^{\infty} e^{-y^2} dy = \frac{e^{a \left(\frac{is}{2a}\right)^2}}{\sqrt{a}} \cdot \sqrt{\pi} = \frac{e^{-s^2/4a}}{\sqrt{a}} \sqrt{\pi}$$

Verify that beta function identity $\beta(a, b) = \beta(a+1, b) + \beta(a, b+1)$

Solution

R.H.S $\beta(a+1, b) + \beta(a, b+1)$

$$\beta(a+1, b) = \frac{\Gamma(a+1) \Gamma(b)}{\Gamma(a+b+1)}, \quad \beta(a, b+1) = \frac{\Gamma(a) \Gamma(b+1)}{\Gamma(a+b+1)}$$

∴ $\beta(a+1, b) + \beta(a, b+1)$

$$= \frac{\Gamma(a+1) \Gamma(b)}{\Gamma(a+b+1)} + \frac{\Gamma(a) \Gamma(b+1)}{\Gamma(a+b+1)} = \frac{\Gamma(a+1) \Gamma(b) + \Gamma(a) \Gamma(b+1)}{\Gamma(a+b+1)}$$

$$= \frac{a \Gamma(a) \Gamma(b) + \Gamma(a) b \Gamma(b)}{(a+b) \Gamma(a+b)} = \frac{\Gamma(a) \Gamma(b) (a+b)}{(a+b) \Gamma(a+b)}$$

$$= \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$$

$\beta(a, b) = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$	L.H.S = R.H.S
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6- By using the generating function for Hermite polynomial $e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$; show that $H_n'(x) = 2n H_{n-1}(x)$

Solution

We have; $e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$

Diff w.r.t 'x' on b/s we get;

$$2t e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n'(x)}{n!} t^n$$

$$2t \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n = \sum_{n=0}^{\infty} \frac{H_n'(x)}{n!} t^n$$

$$2 \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^{n+1} = \sum_{n=0}^{\infty} \frac{H_n'(x)}{n!} t^n$$

Replacing $n \rightarrow n-1$ on L.H.S

$$2 \sum_{n=0}^{\infty} \frac{H_{n-1}(x)}{(n-1)!} t^n = \sum_{n=0}^{\infty} \frac{H_n'(x)}{n!} t^n$$

Equating to zero coefficient of t^n on b/s;

$$2 \frac{H_{n-1}(x)}{(n-1)!} = \frac{H_n'(x)}{n!}$$

$$\frac{2n H_{n-1}(x)}{n(n-1)!} = \frac{H_n'(x)}{n!}$$

'x'ing $\frac{1}{2}$ 'ing n on L.H.S.

$$\frac{2n H_{n-1}(x)}{n!} = \frac{H_n'(x)}{n!}$$

$$H_n'(x) = 2n H_{n-1}(x)$$

Q#2 Answer the following questions:

1- Find Green's function for the equation $-\frac{d^2 y}{dx^2} - \frac{y}{4} = f(x)$ with a boundary condition $y(0) = y(\pi) = 0$.

Solution

For y_c

$$-y'' - \frac{y}{4} = 0 \Rightarrow y'' + \frac{y}{4} = 0$$

Characteristic equation will be:

$$m^2 + \frac{1}{4} = 0 \Rightarrow m = \pm \frac{1}{2}i$$

$$y_c = C_1 \cos \frac{1}{2}x + C_2 \sin \frac{1}{2}x$$

$$G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W}$$

W

$$y_1(x) = \cos \frac{1}{2}x ; y_2 = \sin \frac{1}{2}x$$

$$W = \begin{vmatrix} \cos \frac{1}{2}x & \sin \frac{1}{2}x \\ -\frac{1}{2}\sin \frac{1}{2}x & \frac{1}{2}\cos \frac{1}{2}x \end{vmatrix} = \frac{1}{2} \cos^2 \frac{1}{2}x + \frac{1}{2} \sin^2 \frac{1}{2}x$$

$$= \frac{1}{2} (1) = \frac{1}{2}$$

$$G(x,t) = \cos \frac{1}{2}t \sin \frac{1}{2}x - \cos \frac{1}{2}x \sin \frac{1}{2}t = 2 \cos \frac{1}{2}t \sin \frac{1}{2}x - \cos \frac{1}{2}x \sin \frac{1}{2}t$$

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$\frac{1}{2}$

$$y_c = c_1 \cos \frac{1}{2}x + c_2 \sin \frac{1}{2}x \Rightarrow y(0) = 0 = c_1 \Rightarrow \boxed{c_1 = 0}$$

$$y_c = c_2 \sin \frac{\pi}{2}$$

$$y_p = \int_0^x G(x,t) f(t) dt + \int_{-\pi}^x G(x,t) f(t) dt$$

$$y_p = 2 \int_0^x (\cos \frac{1}{2}t \sin \frac{1}{2}x - \cos \frac{1}{2}x \sin \frac{1}{2}t) f(t) dt + 2 \int_{-\pi}^x (\cos \frac{1}{2}t \sin \frac{1}{2}x - \cos \frac{1}{2}x \sin \frac{1}{2}t) f(t) dt$$

$$\text{Also: } G(x,t) = 2 \cos \frac{t}{2} \sin \frac{x}{2} - 2 \cos \frac{x}{2} \sin \frac{t}{2}$$

$$G(x,t) = \begin{cases} 2 \cos \frac{t}{2} \sin \frac{x}{2} & 0 \leq t \leq x \\ 2 \cos \frac{x}{2} \sin \frac{t}{2} & t \leq x \leq \pi \end{cases}$$



2. Using the Rodrigues formula $P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx}\right)^n (x^2-1)^n$, show that $P_n(x)$ are orthogonal & that $\int_{-1}^1 P_n(x) P_n(x) dx = \frac{2}{2n+1}$

Solution $m=n$

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} t^n P_n(x) = P_0 + tP_1 + t^2P_2 + t^3P_3 + \dots$$

sq. b/s

$$(1-2xt+t^2)^{-1} = \sum_{n=0}^{\infty} t^{2n} P_n^2(x) + \sum_{m,n=0}^{\infty} 2t^{m+n} P_m P_n$$

$\therefore$ Integrating:

$$\sum_{n=0}^{\infty} t^{2n} \int_{-1}^1 P_n^2 dx + \sum_{m,n=0}^{\infty} 2t^{m+n} \int_{-1}^1 P_m P_n dx = \int_{-1}^1 (1-2xt+t^2)^{-1} dx$$

$$\begin{matrix} 1 & & 1 & & 1 & & \\ & & & & & & \\ & & & & & & \end{matrix} = \int_{-1}^1 \frac{dx}{(1-2xt+t^2)}$$

when $m \neq n$ $\int_{-1}^1 P_m P_n dx = 0$

$$\sum_{n=0}^{\infty} t^{2n} \int_{-1}^1 P_n^2 dx \neq 0 = \log \left| \frac{(1-2xt+t^2)}{-2t} \right|_{-1}^1$$

$$= -\frac{1}{2t} [\log(1-2t+t^2) - \log(1+2t+t^2)]$$

$$= -\frac{1}{2t} [\log(1-t)^2 - \log(1+t)^2]$$

$$= -\frac{1}{2t} 2 [\log(1-t) - \log(1+t)] = \frac{1}{t} [\log(1+t) - \log(1-t)]$$

$$\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots$$

$$\log(1-t) = -t - \frac{t^2}{2} - \frac{t^3}{3} - \frac{t^4}{4} - \dots$$

$$\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots$$

$$\log(1-t) = -t - \frac{t^2}{2} - \frac{t^3}{3} - \frac{t^4}{4} - \dots$$

$$2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \dots$$

$$= \frac{1}{t} \left[2t + \frac{2t^3}{3} + \frac{2t^5}{5} + \dots \right] = \frac{2t}{t} \left[1 + \frac{t^2}{3} + \frac{t^4}{5} + \frac{t^6}{7} + \dots \right]$$

$$\sum_{n=0}^{\infty} t^{2n} \int_0^1 P_n^2 dx = 2 \sum_{n=0}^{\infty} \frac{t^{2n}}{2n+1}$$

Comparing coefficients:

$$\int_0^1 P_n^2 dx = \frac{2}{2n+1}$$



3- Show that $\Gamma(n+1)\Gamma\left(n+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^n} \Gamma(2n+1)$ where n is an integer.

Solution

Proof.- We know that,

$$\Gamma\left(n+\frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \Gamma\left(\frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \sqrt{\pi} = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$

$$\Gamma(n+1) = n! = n(n-1)(n-2)(n-3)(n-4) \cdots 4 \cdot 3 \cdot 2 \cdot 1$$

$$\Rightarrow \Gamma(n+1) = \frac{2^n n(n-1)(n-2)(n-3)(n-4) \cdots 4 \cdot 3 \cdot 2 \cdot 1}{2^n}$$

$$\Rightarrow \Gamma(n+1) = \frac{2n(2n-2)(2n-4)(2n-6)(2n-8) \cdots 8 \cdot 6 \cdot 4 \cdot 2}{2^n}$$

$$\Rightarrow \Gamma(n+1) = \frac{(2n)!!}{2^n} = 2^{-n} (2n)!!$$

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Therefore,

$$\Gamma(n+1)\Gamma\left(n+\frac{1}{2}\right) = 2^{-n} (2n)!! \times \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$

$$\Rightarrow \Gamma(n+1)\Gamma\left(n+\frac{1}{2}\right) = 2^{-2n} (2n)!! \times (2n-1)!! \sqrt{\pi} = 2^{-2n} (2n)! \sqrt{\pi}$$

$$\Rightarrow \Gamma(n+1)\Gamma\left(n+\frac{1}{2}\right) = 2^{-2n} \sqrt{\pi} \Gamma(2n+1)$$

In general, replacing n by z , we can write

$$\Gamma(1+z)\Gamma\left(z+\frac{1}{2}\right) = 2^{-2z} \sqrt{\pi} \Gamma(2z+1)$$

PUACP

By directly differentiating $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$
 show that the recurrence relation $2J_n'(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$.

Solution:

from 3rd recurrence relation:

$$\frac{1}{x} J_n(x) + J_n'(x) = J_{n-1}(x) \rightarrow (A)$$

$$-\frac{1}{x} J_n(x) + J_n'(x) = -J_{n+1}(x) \rightarrow (B)$$

Adding (A) & (B)

$$\frac{1}{x} J_n(x) + J_n'(x) = J_{n-1}(x)$$

$$-\frac{1}{x} J_n(x) + J_n'(x) = -J_{n+1}(x)$$

$$2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$$

$$J_n'(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$$

Ex- The function $f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$

is a symmetrical finite step function. Find fourier cosine transform of $f(x)$.

Solution

Fourier Cosine Transform is given as;

$$F_c(s) = \int_0^{\infty} f(x) \cos(sx) dx$$

$$F_c(s) = \int_0^1 f(x) \cos(sx) dx + \int_1^{\infty} f(x) \cos(sx) dx$$

$$= \int_0^1 1 \cos(sx) dx + 0.$$

$$= \left. \frac{\sin(sx)}{s} \right|_0^1$$

$$= \frac{1}{s} (\sin s - \sin 0) = \frac{1}{s} (\sin s)$$

$$F_c(s) = \frac{1}{s} \sin(s)$$

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